

SOFTWARE REQUIREMENTS SPECIFICATION ·
V1.0

CamChat

A real-time mobile
messaging **system**,
engineered for Cameroon.

Real-time chat

24h Status

Voice & Video

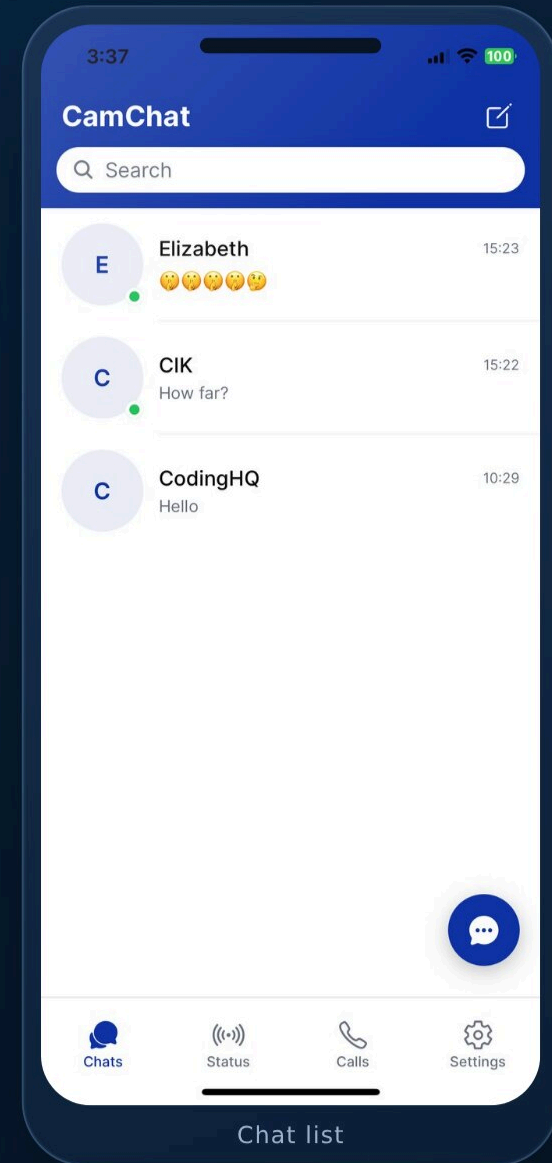
Bilingual EN/FR

Dark mode

Mobile Application Development · Institut Universitaire des Grandes Écoles des
Tropiques (IUGET)

Instructor: Mr. Smith Wills · Academic Year 2025/2026

React Native · Expo · TypeScript · Firebase · Supabase · Agora



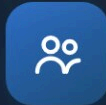
What this presentation covers

A complete walk-through of the CamChat **system** as defined by its SRS — from architecture and actors to functional, interface and non-functional requirements.



System Architecture

BaaS perspective & components



Actors & Use Cases

User classes and interactions



Functional Requirements

10 modules · 70+ requirements



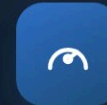
Class & Data Model

UML classes & Firestore design



Interface Requirements

Hardware · software · comms



Non-Functional Reqs

Performance · security · UX

Purpose & product scope

- ✓ Defines the **complete requirements** for CamChat — functionality, interfaces, constraints and quality attributes.
- ✓ A cross-platform app delivering **text, multimedia, voice notes, voice & video calls** and **24-hour status**.
- ✓ Built on a **Backend-as-a-Service** architecture: Firebase, Supabase and Agora.
- ✓ Follows **IEEE Std 830-1998** and **ISO/IEC/IEEE 29148:2018**.

🚩 Localized for Cameroon



+237 default
Country code pre-selected



English & French
Full bilingual interface



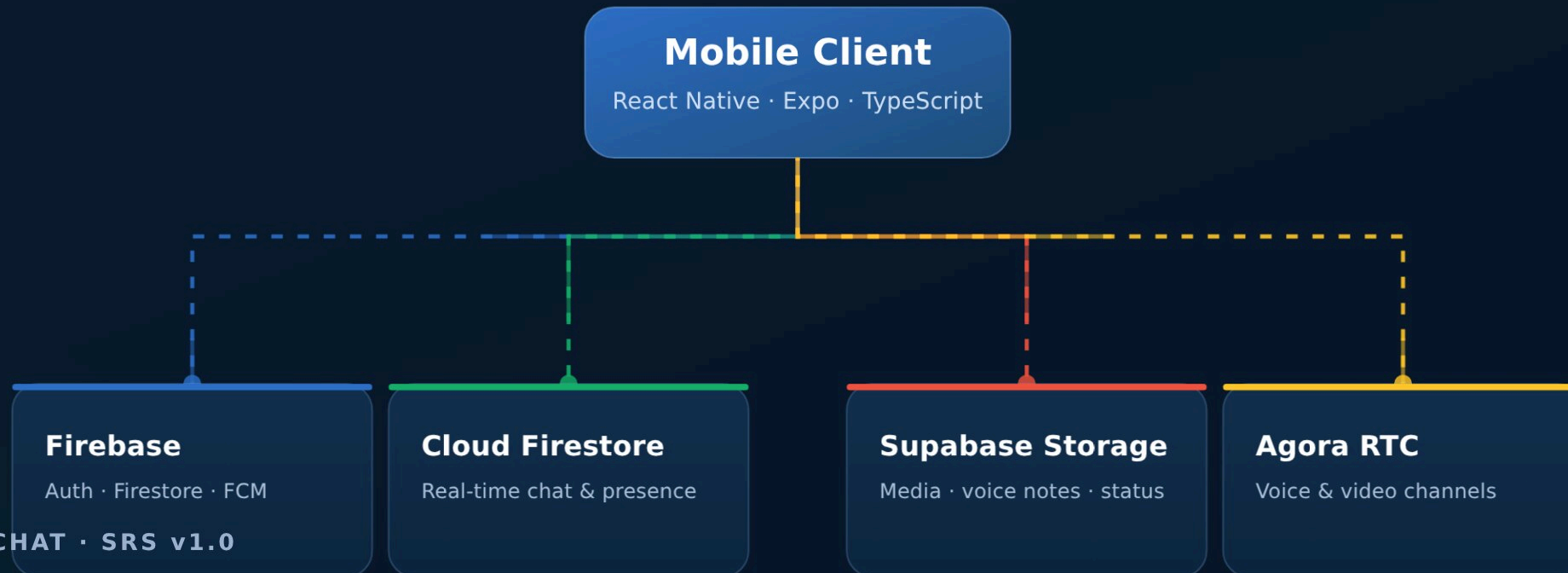
Local identity
Cameroonian onboarding & branding



Globally usable
Accessible beyond Cameroon

System architecture — a Backend-as-a-Service design

A single React Native client communicates directly with managed cloud services — each chosen for what it does best.



Who interacts with the system



Unregistered User

Has installed the app but not registered. Sees onboarding and enters a phone number.

🔍 Onboarding only



Registered User

Completed phone verification & profile setup. Uses all communication features.

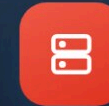
🔍 Full app access



Group Administrator

A registered user with elevated rights inside a specific group conversation.

🔍 + Group management

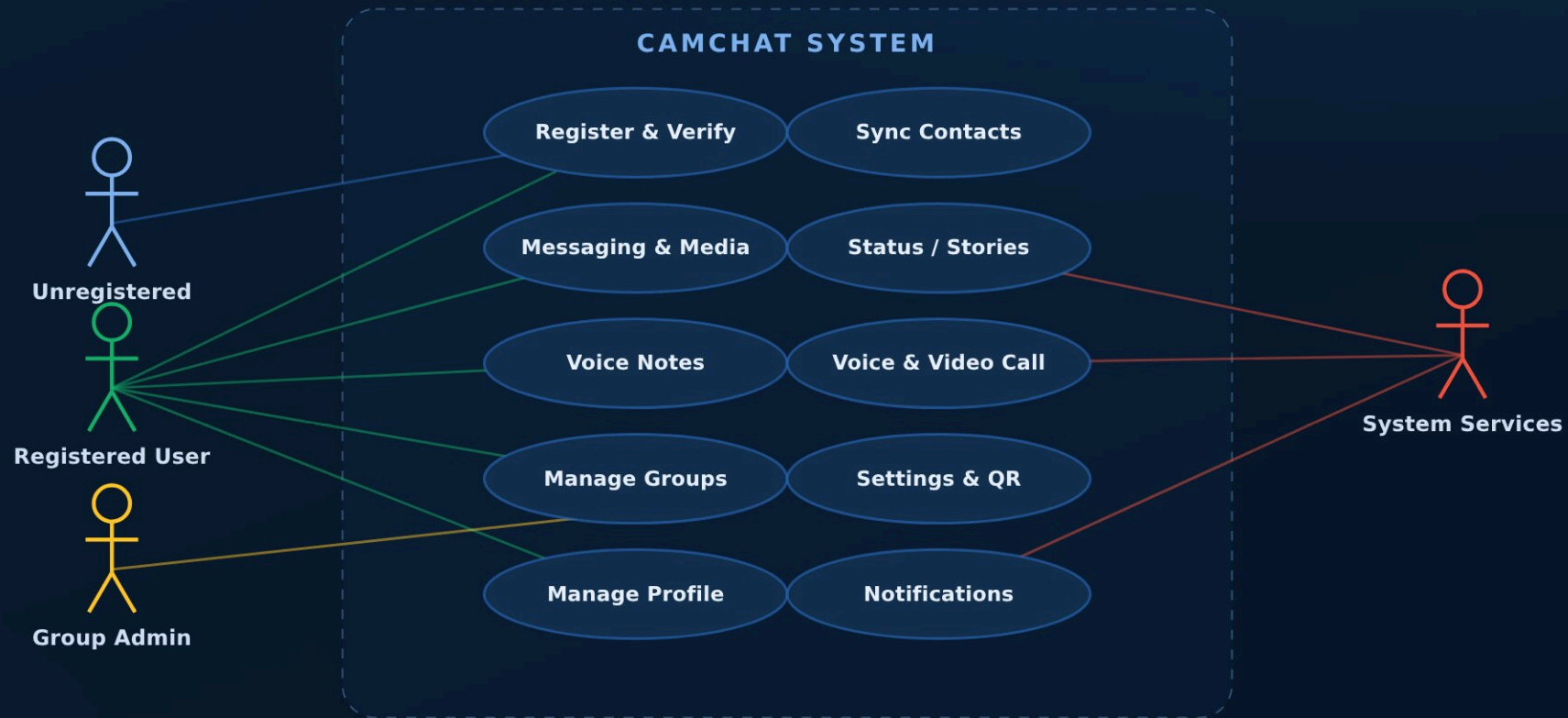


System Services

External cloud services — Firebase, Supabase, Agora, FCM — interacting programmatically.

🔍 Service-level API

How actors interact with CamChat



Ten modules, seventy-plus requirements



Authentication
FR-1 · 21 requirements



Chat & Contacts
FR-2 · 17 requirements



Messaging
FR-3 · 24 requirements



Voice Notes
FR-4 · 7 requirements



Group Chat
FR-5 · 12 requirements



Status / Stories
FR-6 · 8 requirements



Voice & Video
FR-7 · 13 requirements



Notifications
FR-8 · 6 requirements

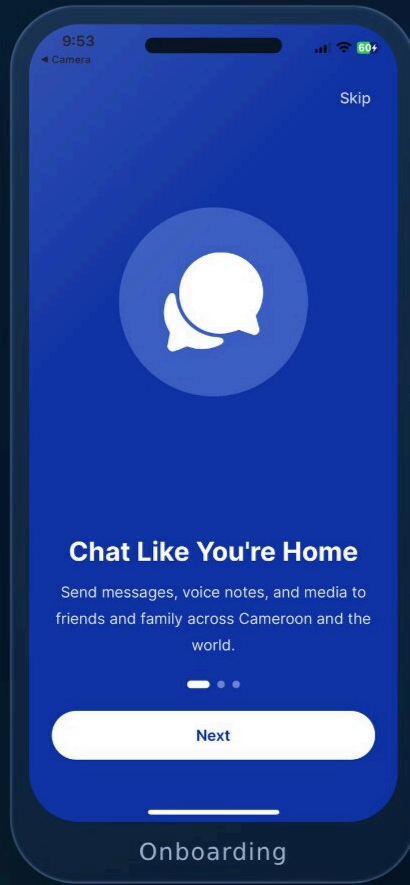


Settings & i18n
FR-9 · 6 requirements

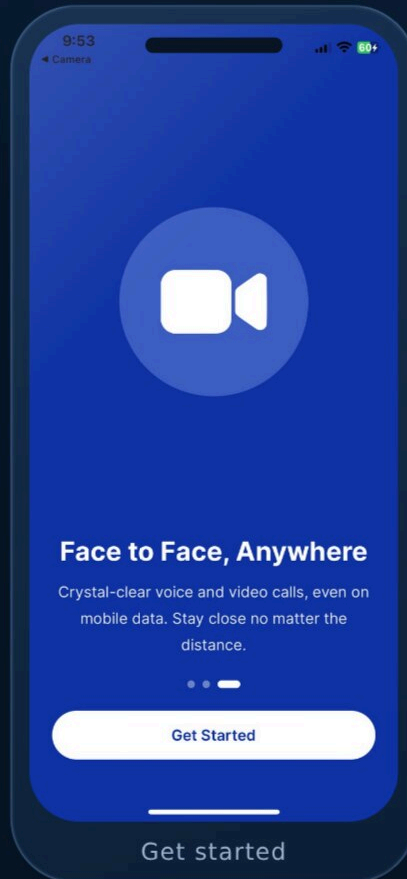


Media Storage
FR-10 · 6 requirements

Phone & OTP — onboarding to first message



Onboarding



Get started

1

Enter phone number

Country selector defaults to +237; validated against E.164.

2

Verify with OTP

One-time password via Firebase Auth; resend cooldown & 10-minute expiry.

3

Set up profile

Display name, avatar and optional About — editable anytime.

4

Persistent session

Stays authenticated across restarts; logout clears the local session.

Rich, real-time conversations



Text & media

Text, images, video, documents & live location.



Voice notes

Record, cancel, send; waveform playback & local cache.



Receipts

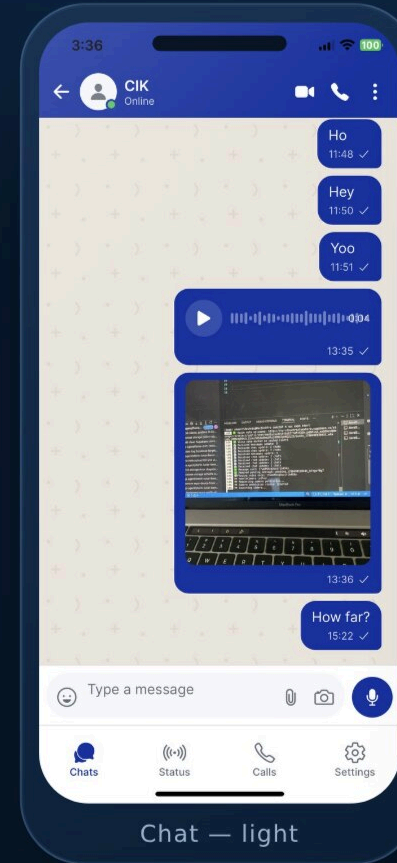
Sending · Sent · Delivered · Read, with timestamps.



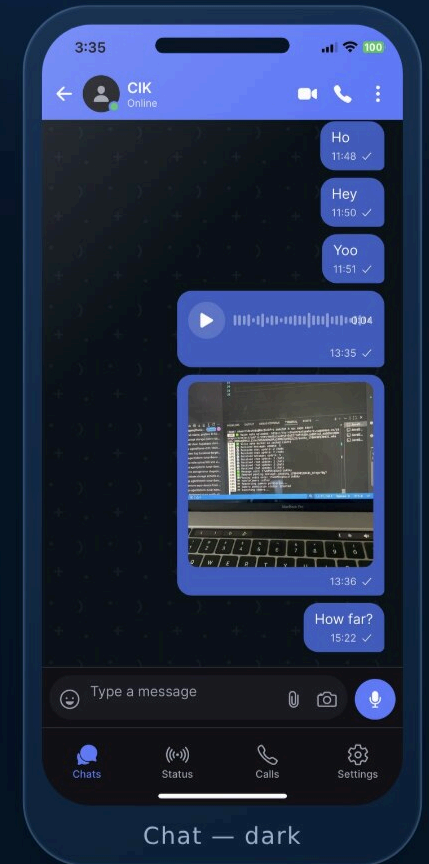
Interactions

Reply, emoji reactions, star & delete; typing indicator.

All media is uploaded to Supabase Storage; the URL is stored in Firestore and streamed back on demand.

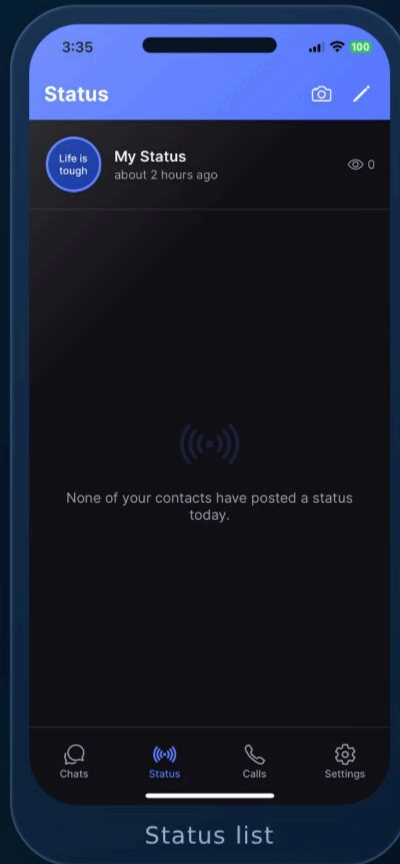


Chat — light



Chat — dark

Communities and 24-hour stories



Status list



Story viewer

Group chat management

- ✓ Create groups with name, image & initial members; creator becomes **admin**.
- ✓ Admins add / remove members, promote or revoke admin rights, edit group info.
- ✓ Any member can view group info and leave at any time.

Status & stories

- ✓ Text, image & video statuses; viewed / unviewed rings & **viewer tracking**.
- ✓ Auto-expire after **24 hours**; creators may delete early.

Real-time communication via Agora RTC



Initiate & receive

Voice or video calls from chat or contacts; accept / decline screen.



In-call controls

Mute, speaker, camera toggle & front/rear switch.



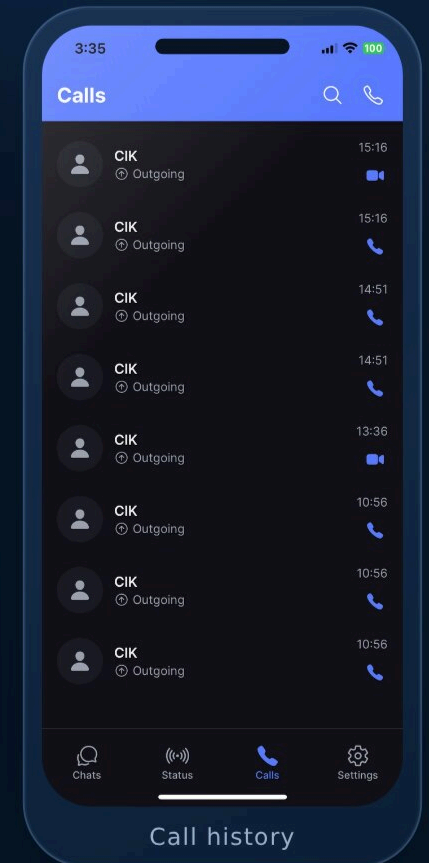
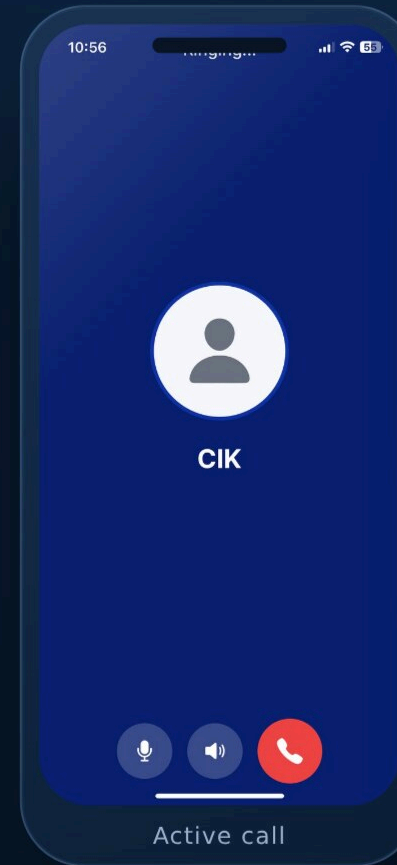
Agora channels

Streams established within 10 s; bitrate adapts to bandwidth.



Call history

Type, direction & duration logged; one-tap return call.



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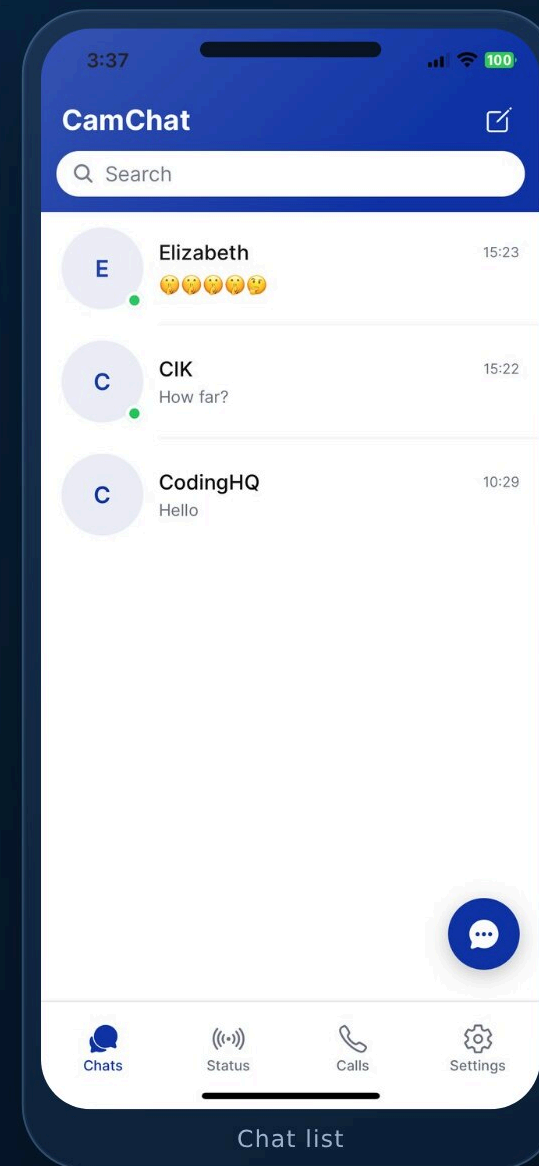
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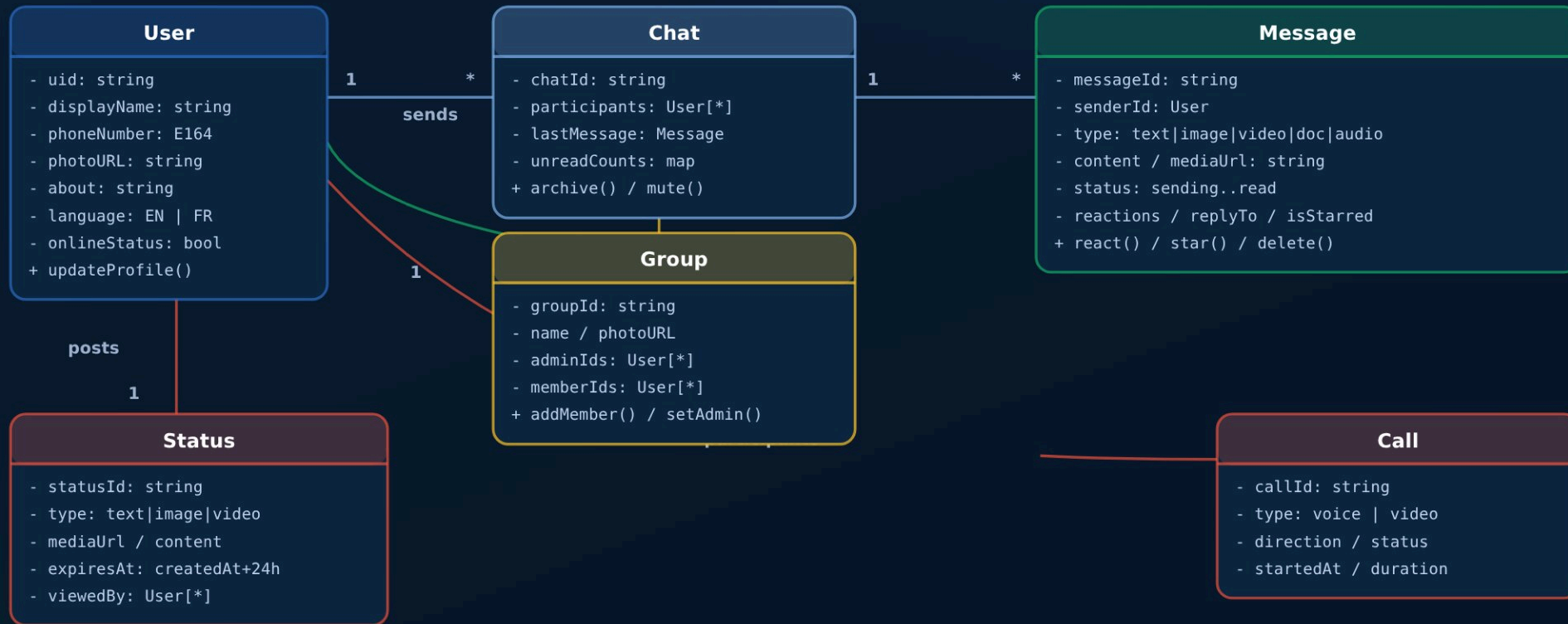
Voice & Video

Bilingual EN/FR

Dark mode



Domain model derived from the data design



Six collections, secured by rules

COLLECTION	DOCUMENT ID	KEY FIELDS	NOTES
users	uid	displayName, phoneNumber, photoURL, about, language, onlineStatus, lastSeen, fcmToken	Created after OTP
chats	chatId	participants[], lastMessage{}, lastMessageAt, unreadCounts{}	Direct & group
chats/{id}/messages	messageId	senderId, type, content, mediaUrl, timestamp, status, reactions{}, replyTo, isStarred, isDeleted	Subcollection
groups	groupId	name, photoURL, adminIds[], memberIds[], createdAt, createdBy	Group metadata
statuses	statusId	userId, type, content, mediaUrl, createdAt, expiresAt, viewedBy[]	Expires +24h
calls	callId	callerId, receiverId, type, status, startedAt, endedAt, duration	Logged on end

Hardware, software & communication

📷 Hardware

- ✓✓ **Camera** — photos, video, calls, QR
- ✓✓ **Microphone** — voice notes & calls
- ✓✓ **GPS** — location messages
- ✓✓ **Contacts** — user discovery

📁 Software

- ✓✓ **Firebase Auth** — phone OTP & sessions
- ✓✓ **Cloud Firestore** — real-time data
- ✓✓ **Supabase** — media storage
- ✓✓ **Agora · FCM · Expo** — calls, push, device

🌐 Communication

- 🔒 **HTTPS / TLS 1.2+** — API & media
- ⚡ **Firestore WebSockets** — live sync
- 📹 **Agora UDP/RTP** — call streams
- 🔔 **FCM / APNs** — push delivery

Performance, security & experience targets

1_s

Text message delivery

3_s

Splash screen on launch

10_s

Call connection

30

Messages per initial page

🛡️ Security

- 🔒 OTP auth before any protected resource
- 🔒 Firestore rules & Supabase policies enforce access
- 🔒 TLS 1.2+ in transit; conversation data is participant-only

👤 Usability

- ✓✓ Register → first message in < 5 minutes
- ✓✓ Core actions in ≤ 2 taps; 44×44 touch targets
- ✓✓ Instant EN/FR switch; WCAG 2.1 AA contrast

🔄 Reliability & scale

- ✓✓ Auto-reconnect & retry of failed messages
- ✓✓ Indexed, paginated queries scale with data
- ✓✓ TypeScript strict · ESLint · feature-based modules

A production-grade, type-safe foundation

</> Frontend

- ✓✓ React Native (Expo)
- ✓✓ TypeScript — strict mode
- ✓✓ expo-router navigation
- ✓✓ Zustand state
- ✓✓ expo-image rendering

Backend & data

- ✓✓ Firebase Authentication
- ✓✓ Cloud Firestore
- ✓✓ Cloud Functions (FCM)
- ✓✓ Supabase Storage
- ✓✓ Agora RTC SDK

🎯 Experience

- ✓✓ i18n — English / French
- ✓✓ Runtime dark mode
- ✓✓ Push notifications
- ✓✓ Skeleton loading states
- ✓✓ Offline-first caching

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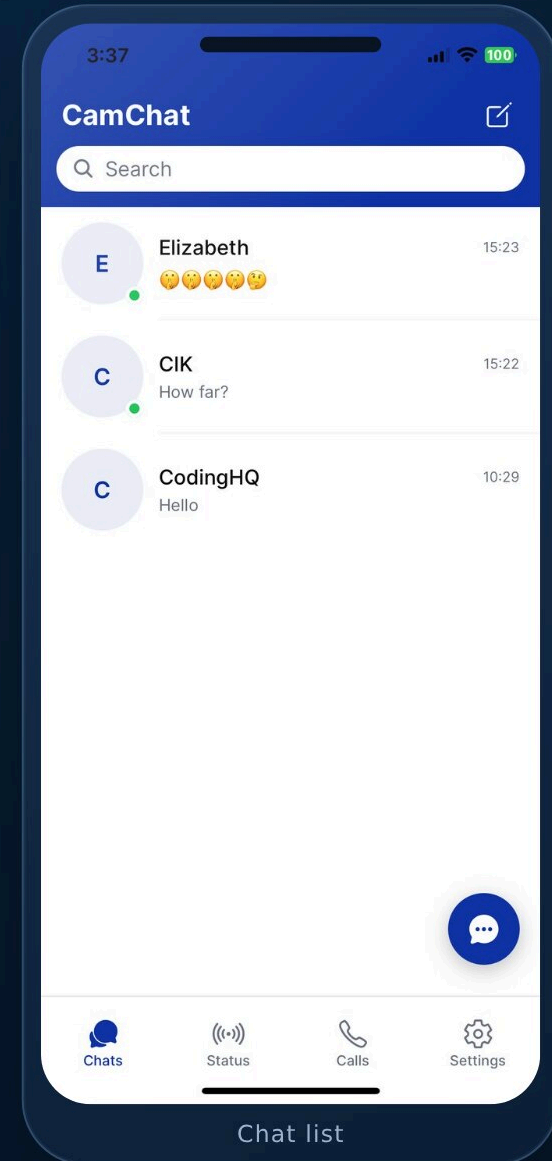
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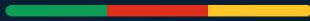
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— THANK YOU

Chat like you're home



CamChat — a secure, reliable, culturally-relevant communication platform for Cameroon and beyond.

👤 Nkwenti Santung Deshnic

👤 Winner Chiyere Madu

👤 Tanyi Georges Nganyuo

👤 Tchakoua Alain Miguel

👤 Chituh Innocentia Kitcha

IUGET · Mobile Application Development · 2025/2026 · Instructor: Mr. Smith Wills